

Zhalok Rahman

Software Engineer 2 at Datatree Technology

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Github: <https://github.com/zhalok>

Video Introduction: <https://www.loom.com/share/c5ab46a4737244edafd9f052555767a1>

Experience

Kinetik Healthcare

Software Engineer 2

2025-04-01 – Present

- Implemented push based realtime system using Redis PubSub
- Generated notifications and pushed in Redis Streams via MongoDB CDC realm trigger
- Implemented RBAC validation and realtime access revoke on role change
- Built endpoints for AI agent tool calls
- Built AI call center agent with ElevenLabs
- Integrated human call center as fallback for AI callcenter service
- Managed ElevenLabs AI agent as code with **Python** SDK, building a CI/CD deploy pipeline that fetches the agent config from the development agent, replicates it into staging/production with injected environment config, and avoids shipping UI-edited agents directly for better auditability
- Built a **Python** voice service (ai-service) with HTTP and WebSocket endpoints: Twilio hits the HTTP endpoint to start a session, receives TwiML pointing to the WebSocket endpoint with the session id, then streams voice over that WebSocket connection
- Streamed incoming voice audio from the WebSocket connection to the LLM orchestration service over gRPC streams
- Integrated Langfuse for tracing and monitoring the voice AI agent pipeline
- Managed infrastructure in AWS ECS
- Debugged build and deployment issues

PropertyFinder

Software Engineer

2024-10-01 – 2026-03-31

- Worked on B2B modernization with Golang and AWS based services like DynamoDB, Lambda, SQS, ECS
- Performed load testing experiment with go script and created a dashboard with necessary metrics to observe the changes in high load
- Implemented asynchronous business logics with SQS, Lambda (go) and Kinesis streams
- Fixed and investigated production incidents
- Collaborated with data engineering team for fixing data discrepancy issues during migration
- Written go scripts for resolving incident and migration of data
- Contributed to data migration using **Python** and Apache Spark scripts
- Fixed github actions deployment failures

Learnyx AI

Fullstack Engineer

2023-10-01 – 2025-02-01

- Built an ai-service entirely in **Python**, consuming RabbitMQ as the entrypoint invocation for heavy AI workloads like vector search and LLM orchestration
- Built AI generation service with Langchain, FastAPI and OpenAI models
- Built websocket and server sent events endpoints for realtime AI generation delivery
- Created RabbitMQ consumers for handling heavy workloads asyncly with PikaClient
- Offloaded event loop blocking sync tasks into background tasks
- Scheduled gathered coroutines for concurrent resolutions
- Traced event loop blocking issues with sync operations and optimized the response time

Innovext IT

Fullstack Developer

2022-12-01 – 2023-10-01

- Built REST apis with Nest JS, Express JS and FastAPI (for deploying fine-tuned model)
- Implemented payment and monetary transaction features by integrating third party payment gateways (stripe and paypal)
- Built client side application with Next JS, used Redux for efficient data fetching and dataflow
- Built an internal monetization system with credits, ensured smooth credit transfer with mongodb sessions
- Built server with FastAPI for inferencing fine-tuned object detection model in onnx format
- Created scripts for data preprocessing and training models

Projects

MongoDB vs PostgreSQL benchmarking for storing documents

- Benchmark results: <https://bit.ly/4dbwS4h>
- Github: <https://bit.ly/3TYSpGZ>

Tech: Golang, JavaScript, Docker, PostgreSQL, MongoDB, Prometheus, Grafana, Loki, Promtail
PostgreSPECTOR

- Demo: <https://youtu.be/w2OCn0X3bRw>
- Github: <https://bit.ly/4y8AFXX>

Tech: Golang, Prometheus, Grafana, PostgreSQL, Loki, Promtail, Docker
Obstainer

- Demo: <https://youtu.be/sifxPdFs2AE>
- Github: <https://bit.ly/4y8AL1L>

Tech: Golang, Grafana, Prometheus, Docker, Loki, Promtail
Data Stores Python

- Benchmark results: <https://bit.ly/3V6UUr9>
- Github: <https://bit.ly/4xMs7q>

Tech: Python, Socket, System Calls, Docker
Remote Browser Control

- Built FastAPI app for receiving commands for playwright to apply on browser session via websocket and emit current snapshot of the browser
- Created a front end app with react to render the remote browser session and also send commands via websocket
- Demo: <https://bit.ly/4wDD7om>
- Github: <https://github.com/zhalok/remote-browser-control>

Tech: Playwright, Python, FastAPI, WebSocket

Invoice Extractor

<https://github.com/zhalok/invoice-extractor.git>

- Built a FastAPI upload/job-polling API and a **Python** RabbitMQ worker pool running an extract, hydrate, verify, and register pipeline
- Designed dual auto-submit and manual-review flows: verified invoices auto-register immediately, while unresolved supplier, duplicate, or math-mismatch cases fall back to a human review UI before submission
- Persisted job and extraction state in Postgres so the UI can poll job status and list every processed job
- Abstracted LLM/OCR extraction behind swappable clients (worker/llm_client.py, worker/ocr_client.py) to decouple the pipeline from the extraction backend
- Containerized the full stack (API, worker, Postgres, RabbitMQ, mock accounting API) with Docker Compose and added a GitHub Actions CI workflow to build and boot the stack on every push/PR
- Github: <https://github.com/zhalok/invoice-extractor.git>
- Demo: <https://youtu.be/BsikeXZiBuI>

Tech: ****Python****, FastAPI, RabbitMQ

Publications

Optical Mark Recognition with Object Detection and Clustering

2024 6th International Conference on Electrical Engineering and Information and Communication Technology (ICEE-ICT)
May 2024

<https://doi.org/10.1109/ICEEICT62016.2024.10534561>

- Written **Python** scripts for generating synthetic training data
- Built training scripts in **Python** with ultralytics for training lightweight YOLO models
- Implemented clustering based unsupervised algorithm for column detection and serialization of answers
- Exported finetuned model in Onnx format for inferencing via API
- demo: <https://www.youtube.com/watch?v=BiEyBE5qQqw>

Certifications

AWS Certified Cloud Practitioner

AWS

<https://www.credly.com/badges/201260d3-12e0-4d22-a49f-4e64d68b348e>

Deep dive on AWS services

Optimizing MongoDB Performance with Tuning Tools

MongoDB

<https://www.credly.com/badges/e861b7b4-028a-47d4-aed6-7e586d4d95da>

Learned about diagnosing and tuning MongoDB performance using database tooling

Supervised Machine Learning: Regression and Classification

Coursera

<https://coursera.org/share/0ef4defbf44ee7d868c52c7e1a3bb94f>

Supervised learning from scratch, learning about mathematical algorithms working as backbone of machine learning models

Education

Shahjalal University of Science and Technology

Bachelors of Science in Computer Science

2019-01-01 – 2024-04-01